Historical Passenger Survival and Risk-Factor Analysis

RMS Titanic Manifest | Excel Capstone Project

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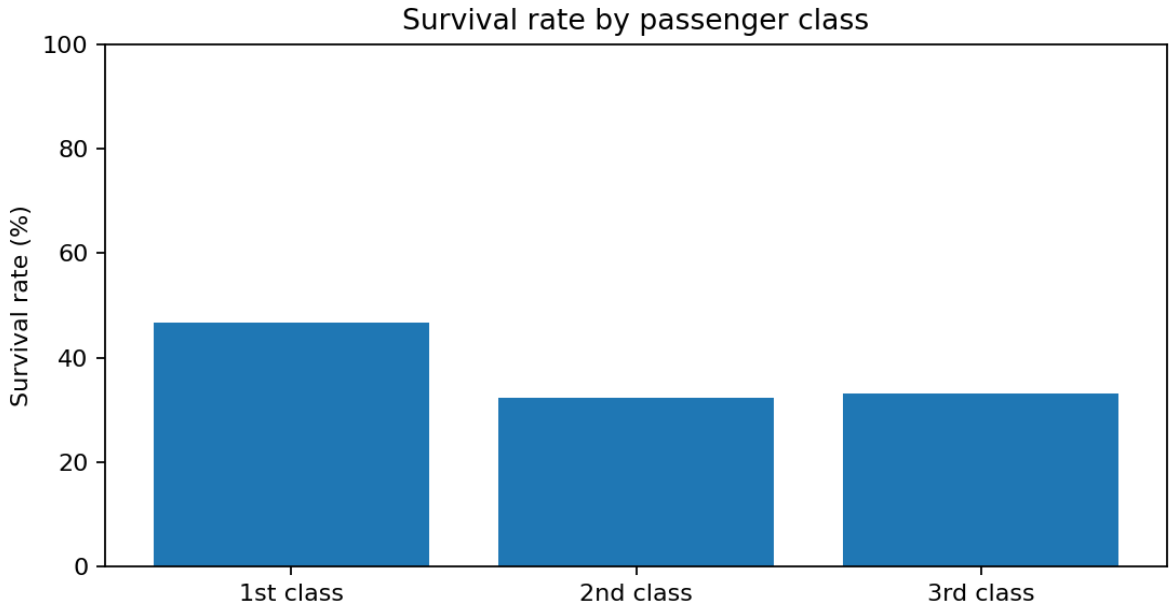
Scope note: This analysis uses only the 418-record Titanic.csv sample supplied for the capstone. It does not represent the full historical Titanic manifest.

1. Executive Summary

The supplied dataset contains 418 passenger records, of which 152 are labelled as survived, giving an overall labelled survival rate of 36.4%. The strongest observed pattern is that the file labels all 152 female records as survived and all 266 male records as not survived. This perfect separation is unusual and must be validated against the instructor/source before drawing historical conclusions.

Class differences are smaller than the sex split in this particular file: first class has a 46.7% labelled survival rate, second class 32.3%, and third class 33.0%. Fare has a weak positive correlation with the binary outcome ($r \approx 0.192$), while age and family size show near-zero and weak positive linear correlations respectively after the documented treatment of missing values. These are associations, not causal effects.

Practical implication: use the workbook to demonstrate a reproducible analytical workflow, but treat the apparent sex-outcome relationship as a data-validation issue first. Do not present it as a generalisable modern passenger-risk model without verifying outcome provenance and testing on a suitable, representative dataset.



2. Problem Statement and Objectives

The business question is: what passenger characteristics are associated with survival outcomes in the supplied sample, and what limitations should a stakeholder consider before using those patterns to inform passenger-safety decisions?

Objectives: (1) profile and document the raw file; (2) apply transparent missing-value and outlier policies; (3) compare outcomes by class, sex, age band, family size, fare and embarkation port; (4) demonstrate lookup and statistical techniques in Excel; and (5) communicate findings with clear caveats.

3. Data Preparation and Methodology

The supplied CSV has 418 rows and 12 source columns. The profile found 86 missing Age values, 327 missing Cabin values, one missing Fare value, and no missing Embarked values. No exact duplicate rows or duplicate PassengerId values were detected.

Missing values. Preserve original Age and Fare values. For analysis only, impute missing age using the median within passenger class and sex, falling back to the overall median if a group has no median. Impute the one missing fare using the median fare for the passenger's class. Retain Cabin because it is highly sparse and add a missingness flag. Embarked requires no imputation because no blanks were found.

Outliers and duplicates. Flag potential Age and Fare outliers using the 1.5×IQR rule, retain them pending review, and document any later exclusions. No duplicate rows or passenger IDs were found. A reconciliation check should confirm clean rows + excluded rows = 418.

Excel implementation note. The attached workbook is a prepared starter containing the source data, analysis-ready fields, documentation, parameters, summaries and dashboard charts. To meet the full rubric, open it in desktop Excel and complete the mandatory Power Query staging/reference queries, named ranges, PivotTables, PivotCharts, Slicers, GETPIVOTDATA KPI and integrity checks. The workbook itself does not claim that these interactive Excel objects have already been created.

4. Analysis and Findings

Segment	Records	Survived	Labelled survival rate
1st Class	107	50	46.7%
2nd Class	93	30	32.3%
3rd Class	218	72	33.0%

Class: first class has the highest labelled survival rate (46.7%), compared with 32.3% in second class and 33.0% in third class. These class comparisons are descriptive and may be affected by the outcome-label pattern noted above.

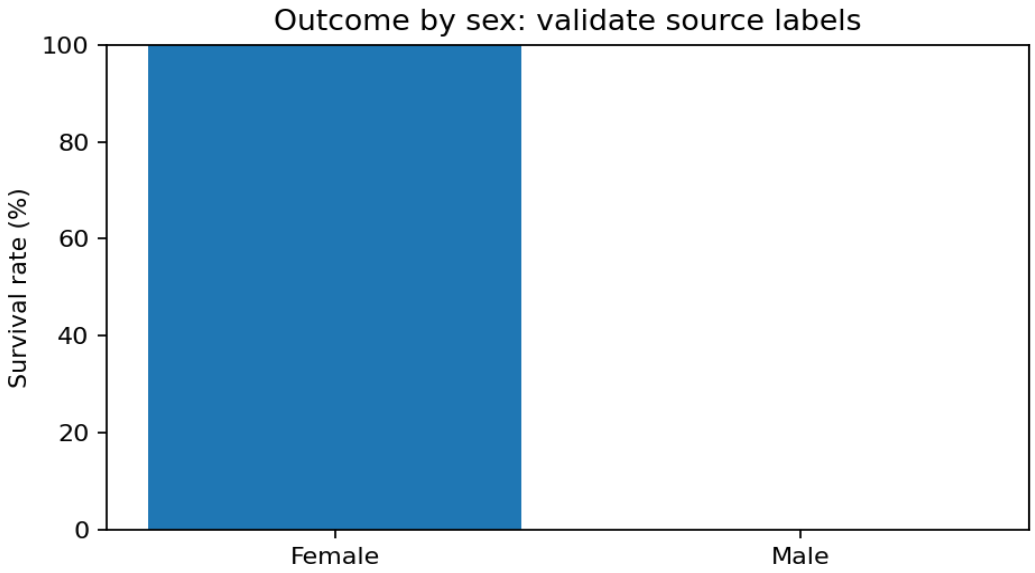
Sex: female records show 100% labelled survival and male records 0%. Because this is a perfect separation, validate the file with the instructor before treating the field as a credible predictor or making a historical claim.

Age: after imputation for grouping, child (0–12) records have a 48.0% labelled survival rate, teen (13–18) 41.4%, adult (19–35) 35.8%, mature (36–60) 39.4%, and senior (60+) 36.4%. The senior band has only 11 records, so its estimate is imprecise.

Family size: survival labels are higher among some small-family groups, but counts become very small at larger family sizes. Avoid ranking large-family segments without showing sample sizes.

Fare: mean analysis fare is approximately 94.28 in first class, 22.20 in second class and 12.44 in third class. The fare/outcome correlation is weakly positive (r ≈ 0.192); class and other variables may confound this association.

Embarkation: the sample's labelled survival rates are 39.2% for C, 52.2% for Q and 32.6% for S. Port Q has only 46 records, and port-level differences should not be interpreted causally.



5. Recommendations

• Validate the source and definition of Survived before making substantive claims. The perfect split by sex is the most important quality concern and may indicate a special construction of the supplied sample.

• Use the workbook's reconciliation and missingness checks at every refresh. Preserve original fields and keep imputed analysis fields separate so transformations remain auditable.

• Show both rates and counts in the dashboard. Suppress or annotate low-count groups, especially larger family sizes and the senior age band.

• Treat correlation and any regression output as exploratory association, not causation. A credible modern safety model would require a representative dataset, validated outcomes, additional operational variables and out-of-sample testing.

6. Conclusion and Further Work

The supplied file supports descriptive comparisons across class, age, family size, fare and embarkation port, but its outcome labels show a perfect relationship with sex. The first next step is source validation, not operationalising the observed pattern. Further work should include verifying how the labels were generated, comparing against an approved reference file if the instructor authorises it, building Power Query refresh steps and interactive PivotTable-based visuals, and testing any proposed model on independent representative data.

7. Appendix: Key Reconciliation and Quality Facts

Check	Result
Raw rows	418
Clean rows retained	418
Excluded rows	0
Reconciliation	418 + 0 = 418
Missing Age	86
Missing Fare	1
Missing Cabin	327
Missing Embarked	0
Exact duplicate rows	0

Tools-use note: Digital tools supported early-stage analysis structuring, formula examples, documentation, and draft preparation. The author is responsible for verifying all outputs and ensuring the final submission reflects their understanding.